

class12: Lab Session

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Section 1. Proportion of G/G in a population

Downloaded a CSV file from Ensemble < https://www.ensembl.org/Homo_sapiens/Variation/Sample?db=core;r=17:39894595-39895595;v=rs8067378;vdb=variation;vf=959672880#373531_tablePanel >

Here we read this CSV file

```
mxl <- read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(mxl)
```

```
## Sample..Male.Female.Unknown. Genotype..forward.strand. Population.s. Father
## 1 NA19648 (F) A|A ALL, AMR, MXL -
## 2 NA19649 (M) G|G ALL, AMR, MXL -
## 3 NA19651 (F) A|A ALL, AMR, MXL -
## 4 NA19652 (M) G|G ALL, AMR, MXL -
## 5 NA19654 (F) G|G ALL, AMR, MXL -
## 6 NA19655 (M) A|G ALL, AMR, MXL -
## Mother
## 1 -
## 2 -
## 3 -
## 4 -
## 5 -
## 6 -
```

```
table(mxl$Genotype..forward.strand.)
```

```
##
## A|A A|G G|A G|G
## 22 21 12 9
```

Q5: What proportion of the Mexican Ancestry in Los Angeles sample population (MXL) are homozygous for the asthma associated SNP (G|G)?

```
table(mx1$Genotype..forward.strand.) / nrow(mx1) * 100
```

```
##
##      A|A      A|G      G|A      G|G
## 34.3750 32.8125 18.7500 14.0625

14.0625%
```

GBR

Now let's look at a different population. I picked the GBR "Great Britain".

```
gbr <- read.csv("373522-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378 (1).csv")
gbr
```

```
##      Sample..Male.Female.Unknown. Genotype..forward.strand. Population.s. Father
## 1      HG00096 (M)      A|A ALL, EUR, GBR      -
## 2      HG00097 (F)      G|A ALL, EUR, GBR      -
## 3      HG00099 (F)      G|G ALL, EUR, GBR      -
## 4      HG00100 (F)      A|A ALL, EUR, GBR      -
## 5      HG00101 (M)      A|A ALL, EUR, GBR      -
## 6      HG00102 (F)      A|A ALL, EUR, GBR      -
## 7      HG00103 (M)      A|G ALL, EUR, GBR      -
## 8      HG00105 (M)      A|A ALL, EUR, GBR      -
## 9      HG00106 (F)      G|A ALL, EUR, GBR      -
## 10     HG00107 (M)      G|G ALL, EUR, GBR      -
## 11     HG00108 (M)      A|A ALL, EUR, GBR      -
## 12     HG00109 (M)      G|G ALL, EUR, GBR      -
## 13     HG00110 (F)      A|G ALL, EUR, GBR      -
## 14     HG00111 (F)      A|A ALL, EUR, GBR      -
## 15     HG00112 (M)      G|G ALL, EUR, GBR      -
## 16     HG00113 (M)      G|G ALL, EUR, GBR      -
## 17     HG00114 (M)      G|A ALL, EUR, GBR      -
## 18     HG00115 (M)      A|G ALL, EUR, GBR      -
## 19     HG00116 (M)      G|G ALL, EUR, GBR      -
## 20     HG00117 (M)      A|A ALL, EUR, GBR      -
## 21     HG00118 (F)      G|G ALL, EUR, GBR      -
## 22     HG00119 (M)      G|A ALL, EUR, GBR      -
## 23     HG00120 (F)      G|G ALL, EUR, GBR      -
## 24     HG00121 (F)      A|G ALL, EUR, GBR      -
## 25     HG00122 (F)      G|G ALL, EUR, GBR      -
## 26     HG00123 (F)      G|A ALL, EUR, GBR      -
## 27     HG00125 (F)      A|G ALL, EUR, GBR      -
## 28     HG00126 (M)      G|G ALL, EUR, GBR      -
## 29     HG00127 (F)      G|A ALL, EUR, GBR      -
## 30     HG00128 (F)      A|G ALL, EUR, GBR      -
## 31     HG00129 (M)      G|G ALL, EUR, GBR      -
## 32     HG00130 (F)      A|G ALL, EUR, GBR      -
```

## 33	HG00131 (M)	G G ALL, EUR, GBR	-
## 34	HG00132 (F)	A A ALL, EUR, GBR	-
## 35	HG00133 (F)	G A ALL, EUR, GBR	-
## 36	HG00136 (M)	G G ALL, EUR, GBR	-
## 37	HG00137 (F)	G A ALL, EUR, GBR	-
## 38	HG00138 (M)	A A ALL, EUR, GBR	-
## 39	HG00139 (M)	G G ALL, EUR, GBR	-
## 40	HG00140 (M)	G A ALL, EUR, GBR	-
## 41	HG00141 (M)	G G ALL, EUR, GBR	-
## 42	HG00142 (M)	G G ALL, EUR, GBR	-
## 43	HG00143 (M)	G A ALL, EUR, GBR	-
## 44	HG00145 (M)	A A ALL, EUR, GBR	-
## 45	HG00146 (F)	A A ALL, EUR, GBR	-
## 46	HG00148 (M)	G A ALL, EUR, GBR	-
## 47	HG00149 (M)	G A ALL, EUR, GBR	-
## 48	HG00150 (F)	G A ALL, EUR, GBR	-
## 49	HG00151 (M)	G A ALL, EUR, GBR	-
## 50	HG00154 (F)	G G ALL, EUR, GBR	-
## 51	HG00155 (M)	A G ALL, EUR, GBR	-
## 52	HG00157 (M)	A A ALL, EUR, GBR	-
## 53	HG00158 (F)	A A ALL, EUR, GBR	-
## 54	HG00159 (M)	A A ALL, EUR, GBR	-
## 55	HG00160 (M)	A A ALL, EUR, GBR	-
## 56	HG00231 (F)	A G ALL, EUR, GBR	-
## 57	HG00232 (F)	G G ALL, EUR, GBR	-
## 58	HG00233 (F)	G G ALL, EUR, GBR	-
## 59	HG00234 (M)	G G ALL, EUR, GBR	-
## 60	HG00235 (F)	A A ALL, EUR, GBR	-
## 61	HG00236 (F)	A A ALL, EUR, GBR	-
## 62	HG00237 (F)	A A ALL, EUR, GBR	-
## 63	HG00238 (F)	G G ALL, EUR, GBR	-
## 64	HG00239 (F)	G A ALL, EUR, GBR	-
## 65	HG00240 (F)	G A ALL, EUR, GBR	-
## 66	HG00242 (M)	G A ALL, EUR, GBR	-
## 67	HG00243 (M)	A G ALL, EUR, GBR	-
## 68	HG00244 (M)	G A ALL, EUR, GBR	-
## 69	HG00245 (F)	A G ALL, EUR, GBR	-
## 70	HG00246 (M)	A G ALL, EUR, GBR	-
## 71	HG00250 (F)	G G ALL, EUR, GBR	-
## 72	HG00251 (M)	G A ALL, EUR, GBR	-
## 73	HG00252 (M)	G A ALL, EUR, GBR	-
## 74	HG00253 (F)	A A ALL, EUR, GBR	-
## 75	HG00254 (F)	A G ALL, EUR, GBR	-
## 76	HG00255 (F)	A G ALL, EUR, GBR	-
## 77	HG00256 (M)	A G ALL, EUR, GBR	-
## 78	HG00257 (F)	G G ALL, EUR, GBR	-
## 79	HG00258 (F)	A A ALL, EUR, GBR	-
## 80	HG00259 (F)	G A ALL, EUR, GBR	-
## 81	HG00260 (M)	G G ALL, EUR, GBR	-
## 82	HG00261 (F)	G G ALL, EUR, GBR	-
## 83	HG00262 (F)	A A ALL, EUR, GBR	-
## 84	HG00263 (F)	G A ALL, EUR, GBR	-
## 85	HG00264 (M)	A G ALL, EUR, GBR	-
## 86	HG00265 (M)	G G ALL, EUR, GBR	-

## 87	HG01334 (M)	A G ALL, EUR, GBR	-
## 88	HG01789 (M)	G A ALL, EUR, GBR	-
## 89	HG01790 (F)	G A ALL, EUR, GBR	-
## 90	HG01791 (M)	A A ALL, EUR, GBR	-
## 91	HG02215 (F)	G G ALL, EUR, GBR	-
##	Mother		
## 1	-		
## 2	-		
## 3	-		
## 4	-		
## 5	-		
## 6	-		
## 7	-		
## 8	-		
## 9	-		
## 10	-		
## 11	-		
## 12	-		
## 13	-		
## 14	-		
## 15	-		
## 16	-		
## 17	-		
## 18	-		
## 19	-		
## 20	-		
## 21	-		
## 22	-		
## 23	-		
## 24	-		
## 25	-		
## 26	-		
## 27	-		
## 28	-		
## 29	-		
## 30	-		
## 31	-		
## 32	-		
## 33	-		
## 34	-		
## 35	-		
## 36	-		
## 37	-		
## 38	-		
## 39	-		
## 40	-		
## 41	-		
## 42	-		
## 43	-		
## 44	-		
## 45	-		
## 46	-		
## 47	-		
## 48	-		

```
## 49 -
## 50 -
## 51 -
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## 81 -
## 82 -
## 83 -
## 84 -
## 85 -
## 86 -
## 87 -
## 88 -
## 89 -
## 90 -
## 91 -
```

Find proportion of G|G

```
round(table(gbr$Genotype..forward.strand.) / nrow(gbr) * 100, 2)
```

```
##
##  A|A  A|G  G|A  G|G
## 25.27 18.68 26.37 29.67
```

This variant is associated with childhood asthma is more frequent in the GBR population than the MKL population.

Let's now dig into this further.

Section 4: Population Scale Analysis [HOMEWORK]

Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

Hint: The `read.table()`, `summary()` and `boxplot()` functions will likely be useful here. There is an example R script online to be used ONLY if you are struggling in vein. Note that you can find the median value from saving the output of the `boxplot()` function to an R object and examining this object. There is also the `medium()` and `summary()` function that you can use to check your understanding.

```
expr <- read.table("rs8067378_ENSG00000172057.6.txt")
head(expr)
```

```
##      sample geno      exp
## 1 HG00367   A/G 28.96038
## 2 NA20768   A/G 20.24449
## 3 HG00361   A/A 31.32628
## 4 HG00135   A/A 34.11169
## 5 NA18870   G/G 18.25141
## 6 NA11993   A/A 32.89721
```

Total:

```
nrow(expr)
```

```
## [1] 462
```

Sample sizes for each:

```
table(expr$geno)
```

```
##
## A/A A/G G/G
## 108 233 121
```

Median Expression levels:

```
colnames(expr)
```

```
## [1] "sample" "geno"   "exp"
```

```
tapply(expr$exp, expr$geno, median)
```

```
##      A/A      A/G      G/G
## 31.24847 25.06486 20.07363
```

There are 108 A|A genotypes, 233 A|G genotypes, and 121 G|G genotypes, thus a total of 462 genotypes. The median expression level is 31.248 for A|A, 25.064 for A|G, and 20.074 for G|G.

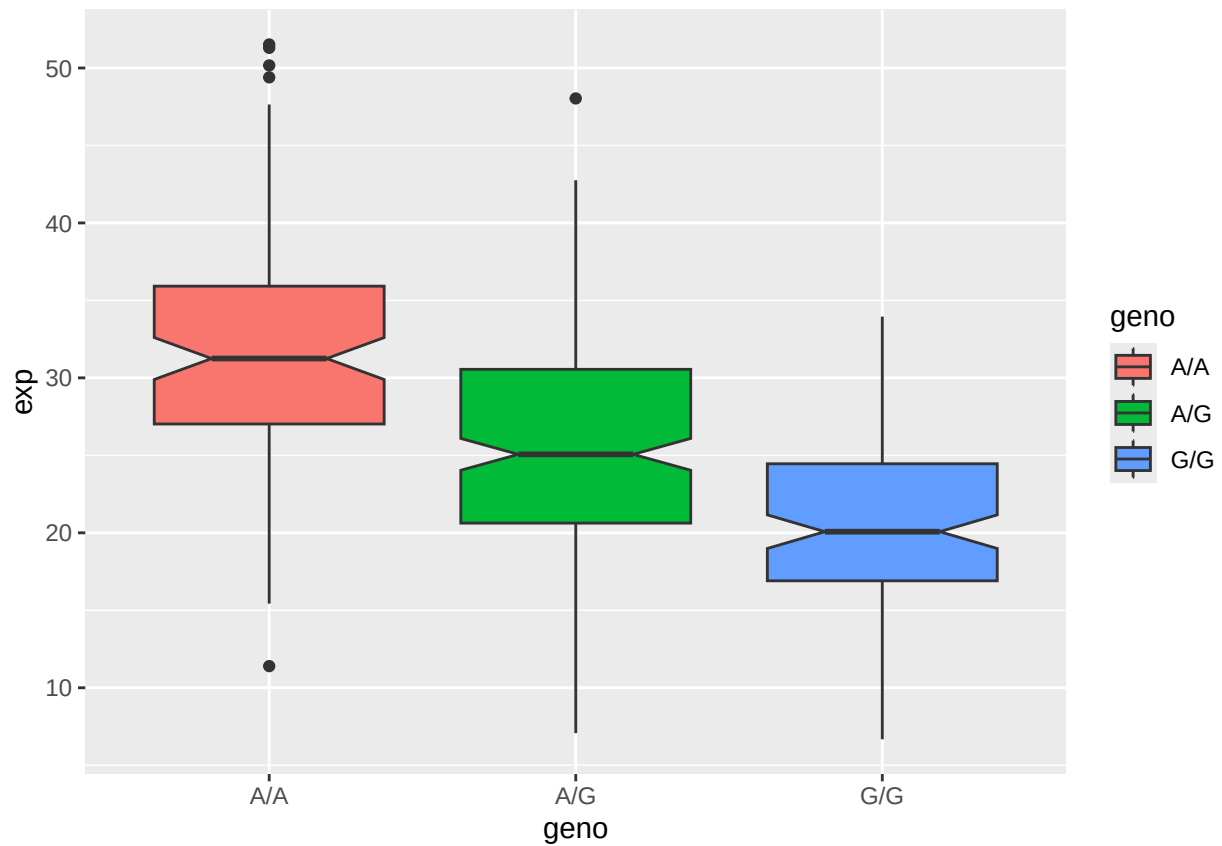
Let's make a boxplot

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
ggplot(expr) +
  aes(x=geno, exp, fill = geno) +
  geom_boxplot(notch=TRUE)
```



From this boxplot, we can infer that the presence of G|G in this gene correlates to a reduction in gene expression. Thus, the SNP affects ORMDL3 expression, indicating a reduction in its expression.